

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

MINI PROJECT

IPL CRICKET DATA ANALYSIS USING PYTHON, POWER BI, AND TABLEAU

Framework for Cricket Data Analytics

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ABSTRACT

The Indian Premier League (IPL) is a professional T20 cricket tournament that generates match, team, player, batting, bowling, and venue-level data. Analysing this information can help identify team performance patterns, scoring trends, player contributions, and factors associated with match outcomes. This mini project presents a workflow for IPL cricket data analysis using Python, Power BI, and Tableau.

Python and Pandas are used to inspect, clean, and transform a match dataset. Matplotlib is used to create charts, while Power BI and Tableau are used to plan interactive dashboards. Example views include wins by team, average runs across season stages, net run rate compared with points, and a dashboard summary of key indicators.

The figures in this report are illustrative demonstration values, not verified official IPL statistics. Results from a real analysis depend on the selected season, data source, column definitions, and data-cleaning decisions.

Keywords: IPL, Cricket Analytics, Data Analysis, Python, Pandas, Power BI, Tableau, Visualization.

Project Objectives

- Load and inspect IPL match data and identify its structure.
- Handle missing values, duplicate records, and inconsistent labels.
- Compare team results and explore scoring and performance patterns.
- Design dashboard views that communicate cricket indicators clearly.

1. DATA PREPROCESSING

Data preprocessing prepares raw match records for reliable analysis. A typical dataset may contain season, match date, venue, teams, toss decision, winner, total runs, wickets, and result margin. Column names should be checked against the file being used.

1.1 Loading and Inspecting the Data

```
import pandas as pd
import numpy as np

df = pd.read_csv("ipl_matches.csv")
print(df.shape)
print(df.info())
print(df.head())
print(df.isnull().sum())
```

The initial inspection shows the number of rows and columns, data types, sample records, and missing-value counts. This helps reveal whether fields such as season and date need conversion and whether numeric measures were imported as text.

1.2 Removing Duplicates and Handling Missing Values

```
df = df.drop_duplicates()

# Standardise text fields
for col in df.select_dtypes(include="object").columns:
    df[col] = df[col].apply(
        lambda x: x.strip() if isinstance(x, str) else x
    )

# Example: fill missing numeric values with median
num_cols = df.select_dtypes(include="number").columns
for col in num_cols:
    df[col] = df[col].fillna(df[col].median())
```

Missing winner values may represent abandoned or tied matches, so they should not automatically be filled with a team name. The treatment must reflect the meaning of each field and the selected dataset.

1. DATA PREPROCESSING (CONTINUED)

1.3 Standardising Categories and Creating Date Features

```
if "date" in df.columns:
    df["date"] = pd.to_datetime(df["date"], errors="coerce")
    df["year"] = df["date"].dt.year
    df["month"] = df["date"].dt.month_name()

# Example derived indicator, if these fields exist
if {"team1", "team2"}.issubset(df.columns):
    df["teams_in_match"] = 2

df.to_csv("Cleaned_IPL_Matches.csv", index=False)
print("Cleaned dataset saved")
```

Date conversion enables season and month comparisons. Team names should be standardised carefully because franchise names can change between seasons. If player-level ball-by-ball data is used, preprocessing may also include extras, legal deliveries, innings, and dismissal-type checks.

1.4 Data Quality Checks

- Confirm that match IDs are unique where expected.
- Check date formats and valid season values.
- Review missing winner, venue, and team fields.
- Verify that runs, wickets, and overs fall within plausible ranges.
- Document exclusions and assumptions before producing charts.

Result: This workflow demonstrates inspection, cleaning, date feature creation, and export. Actual findings depend on the dataset and its source.

2. DATA VISUALIZATION USING PYTHON

Python provides a flexible environment for exploring cricket data. Pandas supports tabular analysis, while Matplotlib can create charts for team comparisons and scoring trends. The following charts use illustrative values only.

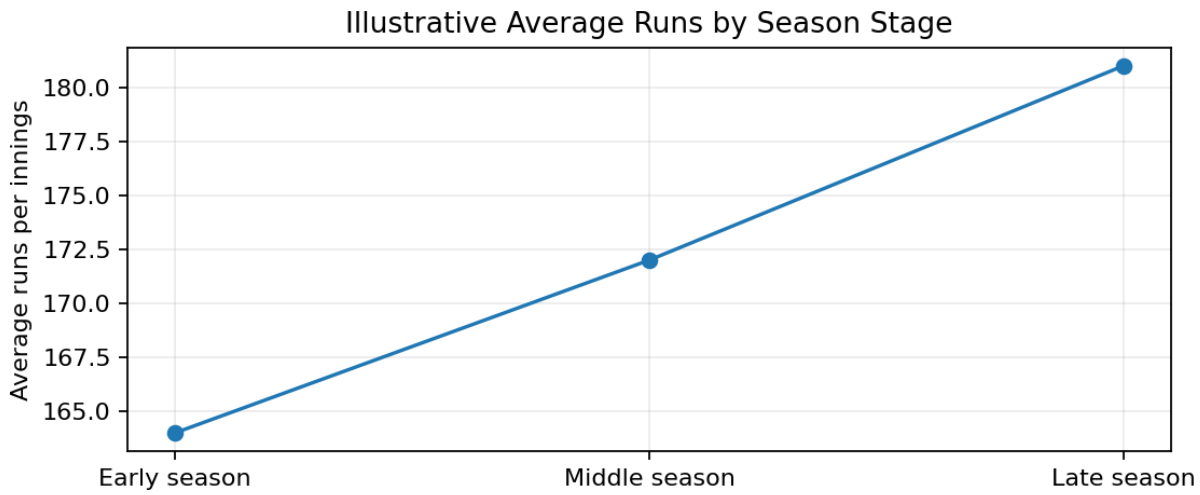


Figure 2.1: Illustrative Average Runs by Season Stage

Interpretation: The example chart shows an upward scoring pattern from early to late season stages. In a real dataset, this pattern should be checked against venue, pitch, innings, team composition, and season effects.

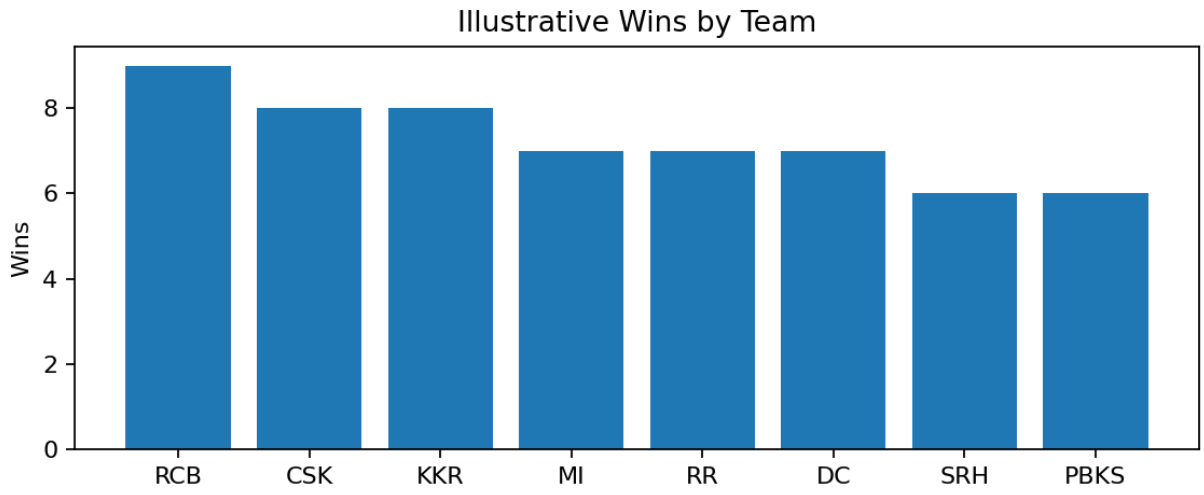


Figure 2.2: Illustrative Wins by Team

Interpretation: A team-wise win chart gives a quick comparison of outcomes. For fair comparison, specify the season and account for matches played, abandoned matches, and tournament format.

2. DATA VISUALIZATION USING PYTHON (CONTINUED)

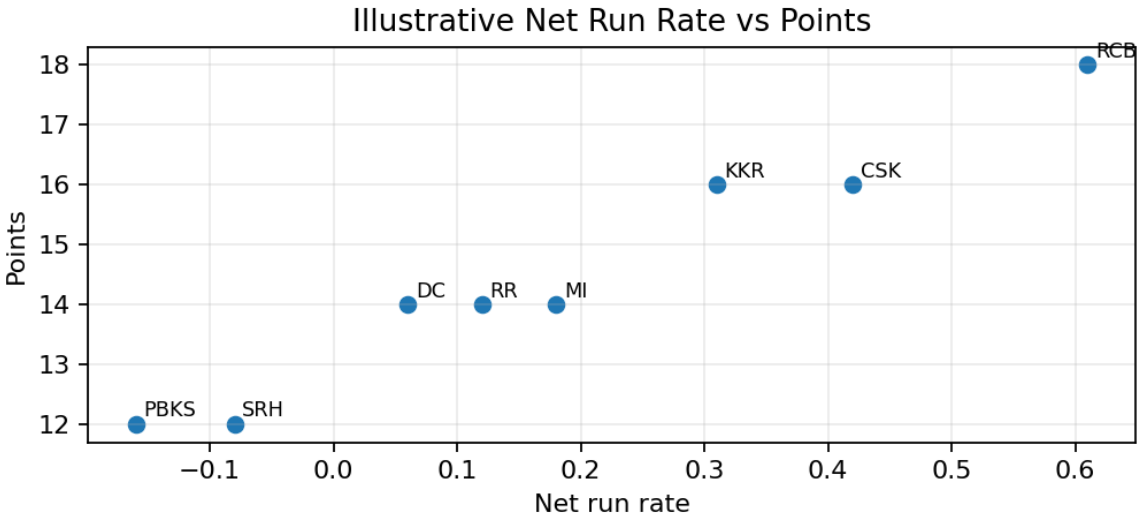


Figure 2.3: Net Run Rate Compared with Points (Illustrative)

Interpretation: The scatter plot explores whether higher net run rate is associated with more points. A relationship in this chart is not proof of causation; the sample values are illustrative and should be replaced with verified season data.

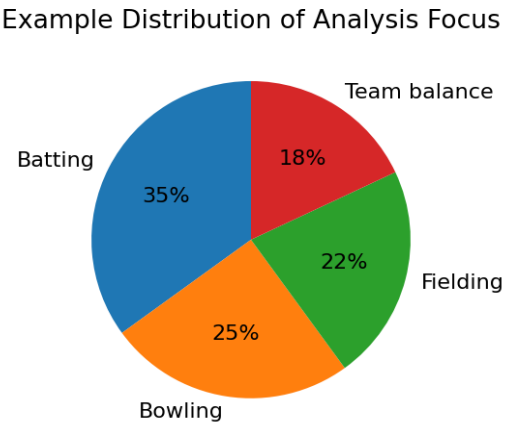


Figure 2.4: Example Distribution of Analysis Focus

The pie chart is a presentation example rather than a measurement from IPL records. For cricket analysis, charts should be based on clearly defined statistics such as runs, wickets, strike rate, economy rate, or win percentage.

3. POWER BI DASHBOARD

Power BI can combine match indicators, team comparisons, and interactive filters in a single report. The dashboard design below describes a proposed layout; it is not a live Power BI file.

Suggested dashboard layout

KPI CARD	KPI CARD	KPI CARD	KPI CARD
Matches	Wins by selected team	Average runs	Highest team total
Team wins chart			
Team filter • Season filter • Venue filter			
Match result table with date, teams, winner and margin			

Figure 3.1: Proposed IPL Performance Dashboard Layout

Dashboard components

- KPI cards: total matches, selected team wins, average runs, and highest total.
- Slicers: season, team, venue, and match stage where supported by the data.
- Charts: wins by team, runs by season, wickets, and result margins.
- Detail table: match date, teams, winner, and outcome margin.

Power BI can use relationships between match, team, player, and delivery tables when the dataset contains these entities. Measures should be defined consistently and tested against source totals.

4. TABLEAU DASHBOARD

Tableau supports visual exploration through worksheets, filters, and dashboards. A proposed IPL dashboard can combine team performance, scoring trends, and match-level details to help users explore different parts of a season.

Suggested Tableau dashboard layout

TEAM PERFORMANCE	SCORING TREND
Wins / losses by team	Average runs by season or match stage
VENUE COMPARISON	**MATCH DETAILS**
Venue-level results and totals	Date • Teams • Winner • Margin
Filters: Season | Team | Venue | Match stage	

Figure 4.1: Proposed IPL Analysis Dashboard Layout

Tableau workflow

- Connect to the cleaned CSV or relational data source.
- Create worksheets for team wins, run trends, venue comparisons, and results.
- Add filters and actions so selecting a team can update related views.
- Assemble the worksheets into a dashboard and validate totals.

The dashboard should use consistent units and labels, show the selected season clearly, and avoid comparing teams across different numbers of matches without context.

5. FINDINGS AND CONCLUSION

5.1 Key Observations from the Demonstration

- Team win charts provide a simple way to compare outcomes within a selected season.
- Scoring trend charts can highlight changes in average runs across a season or match stages.
- Net run rate and points can be viewed together to explore team performance patterns.
- Venue and match-stage filters help users examine results in more specific contexts.
- Interactive dashboards make it easier to move from summary indicators to individual match records.

These are observations about the analytical methods and illustrative charts, not conclusions about any specific IPL season. A final submission should use a named season and a verified dataset to report real findings.

5.2 Conclusion

This mini project demonstrates an IPL cricket data analysis workflow using Python, Power BI, and Tableau. Python supports data inspection, cleaning, transformation, and charting. Power BI and Tableau can present key indicators in interactive dashboards that help compare teams, scoring patterns, venues, and match outcomes.

For evidence-based conclusions, replace illustrative values with verified IPL records, document the source and season, validate derived measures, and refresh the visualizations. Data definitions such as matches played, win percentage, run rate, and player strike rate should be stated clearly.

Tools Used: Python, Pandas, NumPy, Matplotlib, Power BI, and Tableau.

END OF REPORT

APPENDIX: METRICS AND IMPLEMENTATION NOTES

Suggested measures

- **Matches played:** Number of completed matches attributed to a team in the selected season.
- **Win percentage:** (Wins ÷ Matches counted) × 100. State how ties, no-results, and abandoned matches are handled.
- **Average innings runs:** Total runs scored ÷ innings included in the calculation.
- **Net run rate:** A tournament-defined measure based on run-scoring and run-conceding rates; follow the official competition calculation rules.
- **Player strike rate:** (Runs scored ÷ balls faced) × 100, with the dataset's delivery rules applied.

Suggested project folder structure

```
IPL-Cricket-Analysis/
  data/
    ipl_matches.csv
    Cleaned_IPL_Matches.csv
  notebooks/
    ipl_analysis.ipynb
  visuals/
    team_wins.png
    runs_trend.png
  dashboard/
    PowerBI_Report.pbix
    Tableau_Workbook.twbx
  report/
    IPL_Cricket_Data_Analysis_Report.pdf
```

Data and reporting checklist

- Record the data source URL, access date, season, and licence/usage conditions.
- Keep a copy of the raw data and do not overwrite it during cleaning.
- Check chart labels, aggregation logic, and filters before submission.
- Clearly label sample data and avoid presenting illustrative numbers as official statistics.

Note: This appendix provides implementation guidance. The PDF report does not include a connected Power BI or Tableau workbook; those dashboard files must be created separately in the respective applications.